Character accuracy conceals numerical error: silent value corruption in a clinical document pipeline, and what actually prevents it

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Abstract

Background: Clinical decision support systems increasingly read patient data from document images. Accuracy for that stage is reported at character level, and a laboratory value is not a character.
Methods: We measured the image-to-value path of a retrieval-based clinical decision support prototype under controlled image degradation, using synthetic reports carrying exact machine-readable ground truth, with the document as the experimental unit. Four experiments were run: a degradation sweep over 832 readings from one document, a resolution study of 910 readings replicated across 10 reports, a comparison of seven recognition approaches over the same 832 readings plus 1,988 readings from 20 held-out reports typeset in fonts no model had seen, and an evaluation of verification checks over those same extracted values. The 832 readings are shared between the first and third experiments by design, since the third re-reads the first experiment's document with each recognition approach.
Results: At 100 dpi, character accuracy remained at 88.9% while 32.3% of laboratory values were misread. Of 188 corrupted values, 89 (47.3%) carried the same abnormality flag as the truth and so were invisible to the alerting layer. Because flags are threshold quantisations, corruption of an already-abnormal value was more often silent than corruption of a normal one (61.7% against 22.1%; risk ratio 2.80, 95% CI 1.75–4.47), an effect that held with the document as the unit of analysis (paired difference 43.0 percentage points, $p = 0.0078$). Fine-tuning the recogniser made matters worse: it read the document it was tuned for perfectly at low resolution, but against its own base model under identical preprocessing it lost 8.1 percentage points on held-out reports ($p = 0.031$). Upscaling the page before recognition was the largest single gain, improving low-resolution accuracy from 60.0% to 98.5% and the held-out accuracy of the same base model by 11.0 points ($p = 0.0001$). A vision–language model transcribed legible pages accurately but, on degraded pages, produced fluent values that were not printed and moved them toward the normal range. Requiring agreement between two independently built readers caught 193 of 193 corruptions on held-out data at a 19.8% review rate.
Conclusions: Character-level accuracy is not a safety metric for pipelines that reduce measurements to flags, and the errors such pipelines hide fall disproportionately on patients whose values are already abnormal. In the conditions tested, verification by a second independent reader reduced undetected errors more than any improvement to the recogniser itself. All findings come from synthetic documents and bound no claim about performance in deployment.

1 Introduction

A clinical decision support system that reads a patient record from a scanned page performs a chain of inferences before any model reasons about the patient. Pixels become characters, characters become a labelled value, the value is compared against a reference range, and the resulting flag reaches the clinician or the generator. Each step can fail, and the failures do not announce themselves in the same way.

Work on retrieval-augmented generation in healthcare has concentrated on what happens after a value exists: whether the right passage is retrieved, whether the answer is grounded, whether a citation points where it claims to [1, 5, 8]. That literature assumes the number arriving from the document is the number

printed on it. When the document is a photograph or a fax, the assumption is doing work that nobody has measured.

Optical character recognition is normally reported through character error rate, and character error rate treats every symbol as equivalent. A decimal point is one character. In `5.8 mEq/L` it is also the difference between a potassium concentration requiring monitoring and one incompatible with life. A pipeline can therefore post a high character accuracy and still deliver a wrong number, and the number is what the clinical logic consumes.

The clinical stakes are concrete. Laboratory results reach clinicians as scanned or faxed documents in many settings, including transfers between institutions, referrals from primary care, and any workflow where a patient carries paper from one provider to another [11]. A decision support system that reads such a document and returns a treatment suggestion has already made a measurement before it makes an argument. If the measurement is wrong and nothing downstream can tell, the system is confidently wrong in a way its own confidence scores will not reveal.

We state the boundary of this work at the outset. Everything reported here was measured on synthetic reports generated with exact machine-readable ground truth, which is what makes value-level error measurable at all and what makes it impossible to claim a rate for any deployed system. No real patient record was processed, no clinician assessed any output, and no clinical decision or patient outcome was simulated or measured. What follows is a measurement of software behaviour under controlled image degradation, and should be read as establishing the existence and mechanism of a failure mode rather than its prevalence in practice.

We report four experiments on this path. The first sweeps 32 degradation conditions on a single document to establish which corruptions occur. The second takes the document as the experimental unit across 10 independent reports and 7 resolutions, and measures how often values are misread, how often they are lost, and what the system's own flagging layer does with the result. The third asks whether a better reader fixes the problem, comparing seven recognition approaches, including a recogniser we fine-tuned on synthetic laboratory lines, on both the original document and a held-out set of reports typeset in fonts the fine-tuned models never saw. The fourth asks what catches the errors that remain.

Our contributions are the following. We quantify the dissociation between character accuracy and value accuracy in a clinical extraction pipeline. We give a taxonomy of what corruption does to an abnormality flag, and show that the most common outcome is no change at all. We identify the mechanism behind that outcome, which is threshold quantisation, show that it is predicted by whether the true value was already abnormal, and analyse it with the document as the experimental unit so that the effect does not rest on treating readings from one page as independent. We evaluate a physiological plausibility bound and characterise where it systematically fails, and we report that replacing the deterministic extractor with a language model repaired no error and introduced none. We then show that fine-tuning the recogniser is not the remedy it appears to be but an active cost on unseen typography, that a preprocessing change with no learned parameters outperforms it, that a vision–language model substitutes plausible invented values for illegible ones in a direction that moves results toward normality, and that requiring agreement between two independently built readers left no undetected wrong value in either test set.

2 Related work

Retrieval-augmented clinical systems. Retrieval-augmented generation has been proposed widely for clinical question answering, and recent reviews survey its use across healthcare tasks [1, 5]. Systems such as ClinicalRAG retrieve over heterogeneous clinical knowledge [8], and question-answering corpora such as MedQuAD provide evaluation material [2]. These evaluations measure retrieval and generation quality on text that is already text, and assume the number arriving from the document is the number printed on it.

Recognition quality and what depends on it. Tesseract, the engine measured here, recognises text with an LSTM line model operating on segmented lines [10]. Recognition error is conventionally sum-

marised at the character level, and the limitations of that summary are recognised in the document analysis literature: Hemmer et al. observe that recognition errors degrade downstream natural language processing and that some are not resolvable from the image even by a human reader [7]. Work on extraction from scanned clinical reports has argued that accuracy alone is an insufficient criterion, because information lost at the representation layer never reaches the structured output at all [11]. Most directly, Yu et al. benchmarked language-model extraction tools on medical documents and assessed the image distortions commonly introduced by fax machines, finding a significant impact on extraction performance [3]. That result concerns extraction accuracy; the question we take up is what becomes of such an error once the extracted number has been reduced to a flag.

What this literature measures is how recognition quality affects extraction quality. Yu et al. establish that fax-type distortion degrades extraction from medical documents [3]; Wang et al. establish that information absent from the representation cannot be recovered downstream [11]. Both stop at the extracted field.

Abnormality flags and what they hide. A separate literature studies the flags themselves, and studies them from the opposite direction. Kilickaya and colleagues showed that standard reference ranges derived from healthy adults mark too many results abnormal in critically ill patients, producing false positive alerts and alert fatigue, and that customised ranges reduce them [6]. That work treats the flag as over-sensitive. We report the complementary failure: when the value reaching the flag is already wrong, the same threshold structure makes a predictable subset of those errors produce no alert at all. The two findings are consistent, since both follow from a threshold discarding the distance between a value and its boundary.

Our contribution begins one step later. We ask what happens to a numerical error *after* extraction, once the value has been compared against a reference range and reduced to an abnormality flag, and we show that this reduction is itself an information-destroying operation that hides a predictable subset of errors. To our knowledge the specific result, that corruption of an already-abnormal value is several times more likely to be invisible than corruption of a normal one, and the structural reason for it, have not been reported. The second contribution is the comparative one: that the obvious remedy of fine-tuning the recogniser does not generalise, while rescaling and cross-reader agreement do. We are not aware of a prior comparison of these three strategies on the same clinical extraction task with held-out typography.

Generative readers and confabulation. Vision–language models are increasingly used for document understanding, and the risk that a generative reader produces fluent output unsupported by the source is well recognised. In enterprise document extraction, adding a governance layer of validation and consistency checking has been reported to reduce the hallucination rate from 15.7% to 3.2% [9]. Our observation is the clinical specialisation of the same problem: the invented values are physiologically ordinary and biased toward normality, which defeats checks that inspect the value in isolation.

Reporting standards. Reporting guidance for clinical prediction and machine-learning models now asks explicitly for the data pipeline and its failure modes to be described [4]. This study is a measurement of one such pipeline rather than a model development study, and we follow that guidance where it applies by reporting the extraction stage as a source of error in its own right.

3 Methods

3.1 Documents and ground truth

Each synthetic report is generated with a machine-readable truth file recording the exact value, unit, printed reference range and expected flag for 13 analytes: fasting glucose, HbA1c, haemoglobin, ferritin, white cell count, platelets, sodium, potassium, blood urea nitrogen, creatinine, eGFR, troponin I and LDL

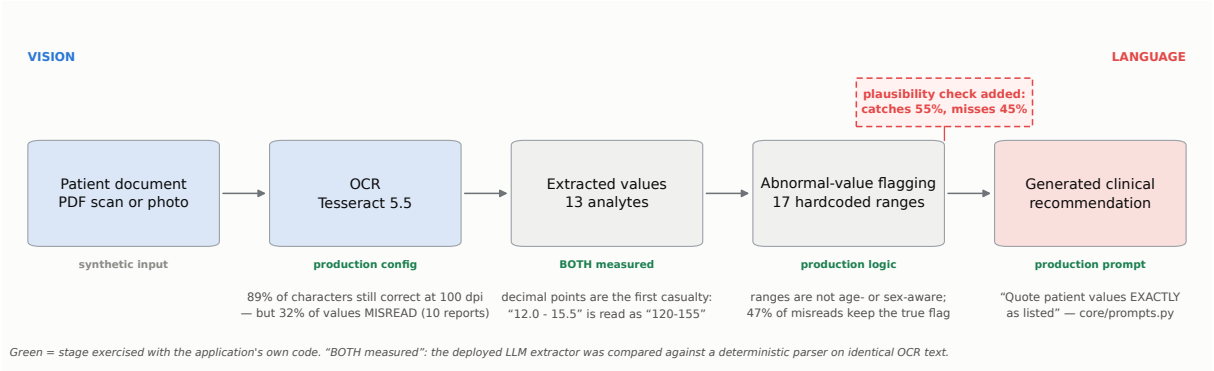


Figure 1: The measured path, with one observed case. A potassium concentration of 5.8 mEq/L printed on the page is extracted as 58.0 and carries the same *High* flag as the true value. Stages marked in green were exercised using the prototype's own code. The potassium case is one observed instance from the single-document sweep and is not a frequency. Whether a corrupted value changes a generated recommendation was not tested.

cholesterol. The panel was chosen to span several orders of magnitude and several decimal conventions, so that a lost decimal point produces an error of different size depending on the analyte.

No real patient data were used at any stage.

3.2 Degradation conditions

Documents are rendered from PDF at a specified resolution and then degraded. The condition set covers resolution (300, 200, 150, 120, 100, 85 and 72 dpi), Gaussian blur ($\sigma \in \{0.5, 1.0, 1.5, 2.0, 3.0\}$), JPEG compression (quality $\in \{90, 70, 50, 30, 15\}$), rotation ($\{0.25, 0.5, 1.0, 2.0, 5.0\}$ degrees), additive noise ($\{5, 10, 20, 30, 50\}$), and five composites intended to resemble real capture paths: a good scan, an office scan, a phone photograph, a poor fax and a poor photograph.

Stochastic conditions are run with five seeds. Experiment 1 therefore comprises 24 single-seed conditions and 8 conditions at five seeds each, giving 64 runs and, across 13 analytes, 832 observations on one document.

3.3 Extraction, flagging and the plausibility bound

Degraded images are passed to Tesseract (`-psm 6`). A deterministic parser locates each analyte by label and extracts the first numeric token on the line.

The flagging logic is transcribed, rather than imported, from the prototype's `clinical_parser` module so that the experiment runs without the application stack. The transcription preserves a property of the deployed system worth stating plainly: its reference ranges are fixed, are not age- or sex-aware, and are not the ranges printed on the document. The system flags haemoglobin outside 12.0–17.5 g/dL while the generated report prints 12.0–15.5 g/dL.

Two properties of this flagging logic matter for interpreting every result below. First, abnormality is decided against *hardcoded* ranges held in the prototype, not against the range printed on the document, and the two can differ: the system flags haemoglobin outside 12.0–17.5 g/dL while the generated report prints 12.0–15.5 g/dL. Second, those ranges are fixed and are neither age- nor sex-aware, and units are assumed rather than parsed, so a change of unit on the page would not be detected. These are properties of the deployed prototype, and we preserve them deliberately rather than correcting them, because the object of study is the pipeline as built. A system using document-printed ranges and parsed units would be exposed to a different, and in some respects smaller, version of the failure we describe.

We additionally implement a physiological plausibility bound, a generous envelope of values possible in a living patient, and ask whether it would have rejected each corruption. This envelope is our own construction for this study, is not a clinical standard, and is not validated against any laboratory's own limits.

Table 1: Experiment 1: outcome by degradation family, 832 observations on one document. *Corrupted* means a wrong value was emitted; *label not found* means the analyte's line was not located at all. JPEG compression to quality 15 and rotation to five degrees produced neither outcome.

Family	Observations	Corrupted (%)	Label not found (%)
Baseline (300 dpi)	13	0.0	0.0
JPEG quality 90–15	65	0.0	0.0
Rotation 0.25–5°	65	0.0	0.0
Noise 5–50	325	1.5	20.0
Blur σ 0.5–3.0	65	9.2	0.0
Composite capture paths	221	6.3	56.1
Resolution 200–72 dpi	78	23.1	10.3

3.4 Experimental unit and the calibration gate

Thirteen analytes read from one page are not thirteen independent observations, since they share a rendering, a scan and an OCR pass. Experiment 2 therefore treats the document as the experimental unit: proportions are computed within each of 10 independent reports and summarised across them, and every dispersion we report is a standard deviation between documents rather than a confidence interval.

No result is reported for a document unless its clean 300 dpi render returns all 13 values correctly. This gate separates degradation effects from generation or parsing faults. All 10 documents passed.

Experiment 2 comprises 10 documents $\times$ 7 resolutions $\times$ 13 analytes, giving 910 observations.

3.5 Outcome definitions

Each reading is *correct*, *misread* (a value was emitted and it differs from truth) or *unread* (no value was emitted for that analyte). Misread and unread are different failures with different consequences and we never pool them.

Character error rate is Levenshtein distance between the whitespace-normalised reference text and the OCR output, divided by the length of the reference.

4 Results

4.1 Which degradations corrupt values, and which destroy them

Experiment 1 sweeps 32 conditions on one document to establish which failure modes exist before committing the replicated design to a single axis. Table 1 gives the outcome by degradation family.

The families separate by failure mode. Resolution is the dominant source of wrong values. Noise and the composite capture paths mostly prevent the analyte's label from being located, so the value is absent rather than wrong. JPEG compression and small rotations produced no error of either kind at the levels tested.

This distinction matters for risk. An analyte that is not found is an absence a caller can detect. A misread analyte is a confident wrong answer. Because resolution is the condition that most readily produces the second case, Experiment 2 replicates the resolution axis across 10 independent documents.

4.2 Character accuracy and value accuracy separate

Table 2 gives both measures across the resolution sweep. At 300 dpi no value was misread in any of the 130 analyte readings. Degradation then affects the two measures at different rates. Between 300 dpi and 100 dpi, character accuracy falls by 9.2 percentage points while the misread rate rises from zero to 32.3%.

Table 2: Character accuracy and value outcomes across the resolution sweep. Each cell is the mean over 10 independent documents, with the standard deviation between documents in parentheses. Misread means a wrong value was emitted; unread means no value was emitted.

Resolution	Character accuracy (%)	Values misread (%)	Values unread (%)
300 dpi	98.10	0.00	0.00
200 dpi	98.02	1.54 (3.24)	0.00
150 dpi	96.67	4.62 (5.38)	0.00
120 dpi	96.12	12.31 (9.73)	0.00
100 dpi	88.88	32.31 (9.46)	10.00 (5.19)
85 dpi	80.83	49.23 (13.66)	14.62 (5.68)
72 dpi	56.53	44.62 (21.08)	41.54 (27.43)

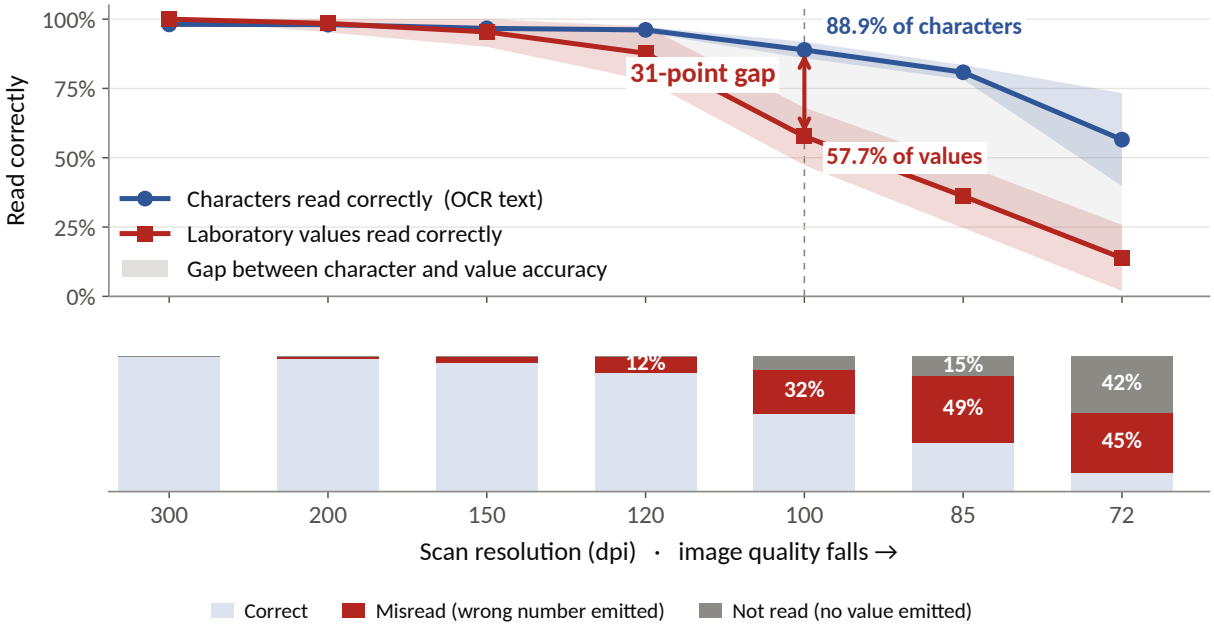


Figure 2: Character accuracy and value accuracy across the resolution sweep. Each point is the mean over 10 independent documents and bands are ± 1 standard deviation between documents, not confidence intervals. The lower panel decomposes each resolution into values read correctly, values misread, and values not read. At 100 dpi the two measures differ by 31 percentage points.

Below 85 dpi the misread rate stops rising while the unread rate more than doubles. Paired by document between 85 dpi and 72 dpi, the misread rate fell for four documents, rose for three and was unchanged for three (mean change -4.6 percentage points, SD 20.9). The data are consistent with values ceasing to be read rather than being read wrongly, and they do not establish a decrease.

4.3 What corruption does to the flag

Across Experiment 2, 188 of 910 readings were misread. Table 3 classifies each by what happened to the abnormality flag. The largest category is no change: 89 corruptions (47.3%) produced a value that carried the same flag as the truth. For those readings the error is invisible to any component that consumes the flag rather than the value.

4.4 Already-abnormal values hide their own corruption

The silent category is not distributed at random. Splitting the 188 corruptions by whether the true value was already outside its reference range gives 61.7% silent (95% CI 52.7–69.9) among the 120 corruptions

Table 3: What corruption did to the abnormality flag, for all 188 misread values in Experiment 2. Categories are mutually exclusive. The silent share computed within each document rather than pooled gives a mean of 48.2 percent (SD 12.7, range 23 to 64 across the 10 documents).

Outcome	Count	Share of corruptions (%)
Flag unchanged (silent)	89	47.3
Normal value flagged abnormal	53	28.2
Flag changed, wrong direction or severity	41	21.8
Abnormal value read as normal	5	2.7
Total	188	100.0

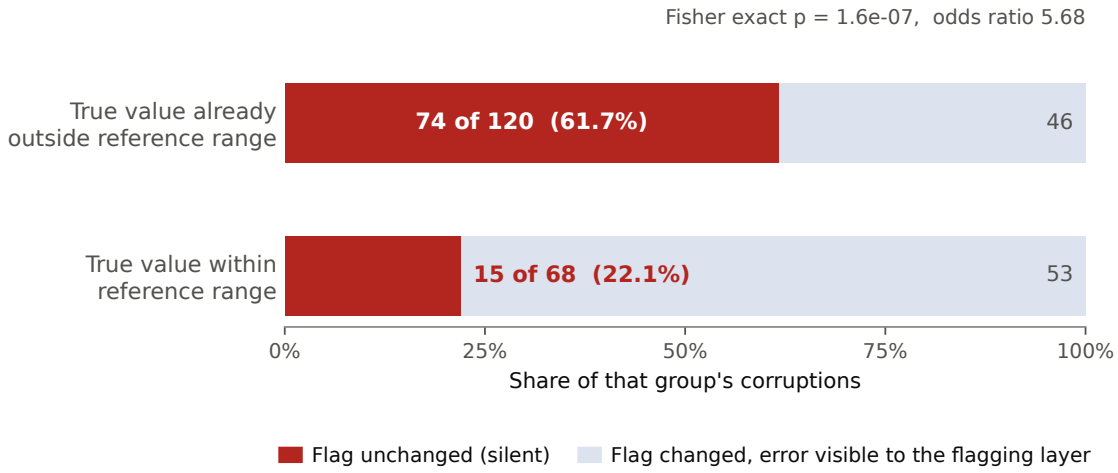


Figure 3: Corruption of a value that was already outside its reference range is far more likely to leave the abnormality flag unchanged. Bars show, within each group, the share of that group's corruptions that were silent. All 188 corruptions from Experiment 2 are included and the two groups are disjoint.

of abnormal values, against 22.1% (95% CI 13.8–33.3) among the 68 corruptions of normal values (Figure 3; Fisher exact $p = 1.6 \times 10^{-7}$). Because corruption is a common outcome in both groups, the odds ratio of 5.68 overstates the multiplicative effect and we report the risk ratio instead: corruption of an already-abnormal value was 2.80 times as likely to be silent (95% CI 1.75–4.47), a risk difference of 39.6 percentage points.

That test assumes the 188 corruptions are independent, and they are not: they are clustered within 10 documents and 13 analytes, and corruptions sharing a document also share a rendering and an OCR pass. We therefore repeat the comparison with the document as the unit of analysis. Eight of the 10 documents produced corruptions of both abnormal and normal values. Within those eight, the silent share is 65.3% (SD 17.3) for corruptions of abnormal values against 22.3% (SD 20.1) for corruptions of normal values, a paired difference of 43.0 percentage points (SD 24.3; Wilcoxon signed-rank $p = 0.0078$; 95% bootstrap CI over documents 26.8 to 58.0 percentage points). The direction holds in all eight documents.

The clustered analysis is the one we rely on. It uses less of the data and gives a wider interval, and it reaches the same conclusion with an effect size that does not depend on treating readings from the same page as independent.

The mechanism is structural rather than incidental. An abnormality flag is a threshold quantisation of a continuous measurement. Once a value lies outside its reference range, moving it further outside does not change which side of the threshold it falls on. Corruption that crosses a threshold is detected by the flagging layer; corruption that drives an already abnormal value deeper into abnormality is invisible by construction.

Decimal-shift errors make the same point quantitatively. Of 188 corruptions, 102 (54.3%) were decimal shifts, all of them upward. Tenfold shifts ($n = 87$) were silent in 42.5% of cases, while hundredfold

Table 4: Plausibility-bound detection rate per analyte, over the corruptions observed for that analyte in Experiment 2. Analytes whose plausible envelope is wide relative to their reference range, or whose envelope is open at the relevant end, are not protected.

Analyte	Corruptions	Detected (%)
Potassium	32	96.9
Haemoglobin	33	90.9
Sodium	6	83.3
Platelets	16	75.0
HbA1c	4	75.0
Creatinine	27	59.3
Blood urea nitrogen	7	42.9
White cell count	18	16.7
LDL cholesterol	8	12.5
eGFR	8	0.0
Fasting glucose	8	0.0
Ferritin	6	0.0
Troponin I	15	0.0

shifts ($n = 15$) were silent in 86.7%. The larger error is the more invisible one.

Troponin I illustrates the consequence. Fourteen of its fifteen corrupted readings were hundredfold overstatements, and every one retained the *Critical* flag: 0.07 ng/mL read as 7.0, 0.82 read as 82.0. A borderline result and a result two orders of magnitude higher are indistinguishable downstream.

4.5 A plausibility bound helps, and fails systematically

The plausibility bound rejected 104 of the 188 corruptions (55.3%) and rejected none of the 636 correct values. The absence of false positives is guaranteed by the generator, which samples truth values strictly inside the envelope, and is not an estimate of a real false-positive rate.

Of the 89 silent corruptions, the bound recovered 33 (37.1%), leaving 56 that neither the flagging layer nor the plausibility check would surface.

Per-analyte detection ranges from 96.9% for potassium to zero for troponin I, ferritin, fasting glucose and eGFR (Table 4). We tested whether detection is predicted by the ratio of plausible envelope width to reference range width. The association is in the expected direction and does not reach significance at this sample size (Spearman $\rho = -0.48$, $p = 0.09$, $n = 13$), and eGFR contradicts it: its envelope is the narrowest of the panel relative to its reference range, yet no corruption was detected, because the envelope's lower bound is zero and every observed eGFR corruption moved the value downward. Detectability depends on where a corrupted value lands relative to an envelope boundary, which is not captured by width alone.

4.6 A language model extractor repaired nothing

We replaced the deterministic parser with a language model extractor (Llama 3.2 3B via Ollama, temperature 0) and ran both over identical OCR text across 12 conditions at two repeats. In four of the 24 runs the model returned no values at all after emitting malformed JSON (`rotate_5.0` twice, `noise_30` twice). Over the 260 readings from the 20 runs that returned values, both extractors were correct on the same 188: zero repairs and zero new errors.

The corruption is established before either extractor sees the text, and a model reading the same damaged characters reproduces the same damaged number.

5 Experiment 3: can a better reader fix it?

Experiments 1 and 2 establish that the corruption happens and that the flagging layer cannot see it. The obvious response is to read the page better. Experiment 3 tests that response directly, including the approach a machine learning practitioner would reach for first, which is to fine-tune the recogniser on in-domain data.

5.1 Test sets

Test A is the same document and the same 64 degradation runs as Experiment 1, giving 832 readings per recognition approach. It is deliberately not new material: it measures each reader on exactly the conditions under which the original failure was characterised. Because it reuses Experiment 1's observations, the two experiments' totals must not be added together.
Test B is held out. It comprises 20 further synthetic reports, 1,988 values, 12 to 18 analytes each, drawn across four column layouts and four laboratory identities, and rendered in **Times and Courier**. Neither font appears anywhere in the fine-tuning data. Each report is degraded under seven conditions: 300, 150, 120 and 100 dpi, an office scan, a phone photograph and a poor fax. Test B exists to detect a model that has learned the study document rather than the task.

The fine-tuning corpus is generated with random values, so no test value can have been memorised.

5.2 Readers compared

- **Stock Tesseract 5.5** (`tessdata_fast`), the pipeline measured above.
- **Tesseract best** (`tessdata_best`, float LSTM), the base for fine-tuning.
- **Rescale variants** of both, in which the page is upscaled to at least 300 dpi by Lanczos resampling before recognition, with the source resolution estimated from page width.
- **Fine-tuned v1** (`labnum`): `tessdata_best` fine-tuned on 6,000 synthetic laboratory lines in sans-serif faces, rendered across 72–300 dpi with blur, noise, JPEG compression and rotation.
- **Fine-tuned v2** (`labnum2`): the same recipe on 8,000 lines spanning sans-serif, serif and monospace families, still excluding Times, Courier and Georgia.
- **PP-OCR** via RapidOCR and ONNX Runtime. Both its detector and its recogniser are unrelated to Tesseract, so it serves as an independent second reader.
- **Qwen2.5-VL 3B** through Ollama, prompted for verbatim transcription, run on a subset of conditions.

Fine-tuning follows the standard `tesstrain` recipe. Training lines are generated programmatically as single-row laboratory entries with the analyte name, a sampled value, its unit and its printed reference range, rendered at a resolution drawn from 72–300 dpi and degraded with the same blur, noise, JPEG and rotation operators used in the evaluation conditions. Version 1 used 6,000 lines in sans-serif faces only; version 2 used 8,000 lines spanning sans-serif, serif and monospace families, excluding Times, Courier and Georgia so that the held-out faces remain unseen. Values are sampled at random within each analyte's plausible envelope, so no specific test value can have been memorised, though the analyte catalogue, reference ranges and number formatting are shared with the test generator (see Limitations).

Line images with word-string box files are converted to `.lstmf` training records, and `lstmtraining` then continues from the stock English LSTM at a learning rate of 10^{-4} for 10,000 iterations before stopping. The base model is `tessdata_best`. No early stopping or validation-based checkpoint selection was used; the final iteration is the released model. The resulting model is 15 MB and runs at the same speed as the stock float model. Recognition uses page segmentation mode 6 throughout, matching Experiments 1 and 2, and the training logs for both versions are released with the code.

Table 5: Exact value accuracy on the 20 held-out reports of Test B, computed per document and summarised across them, with the pooled low-resolution figure from Test A for reference. Both fine-tuned models are built on `tessdata_best` and are evaluated with rescaling, so the comparison that isolates the effect of fine-tuning is against *Tesseract best + rescale* rather than against the stock model. *Silent* is the share of that reader's corruptions that preserved the abnormality flag, which is the quantity the flagging layer cannot see. The low-resolution column is not reported for v1, which was re-evaluated on Test B only.

Reader	Test B accuracy (mean of 20 docs)	SD between documents	≤150 dpi (Test A)	Corrupted (of 1,988)	Silent (% of corruptions)
Stock Tesseract 5.5	74.9%	5.7	60.0%	193	31.1%
Tesseract best	73.0%	6.3	41.5%	204	26.5%
Fine-tuned v1 + rescale	65.9%	14.6	—	482	**59.8%**
Fine-tuned v2 + rescale	75.8%	10.5	**100%**	268	37.7%
PP-OCR (RapidOCR)	78.0%	3.4	95.4%	**58**	69.0%
Stock + rescale	81.4%	2.7	98.5%	121	33.9%
Tesseract best + rescale	**83.9%**	**1.4**	92.3%	89	32.6%

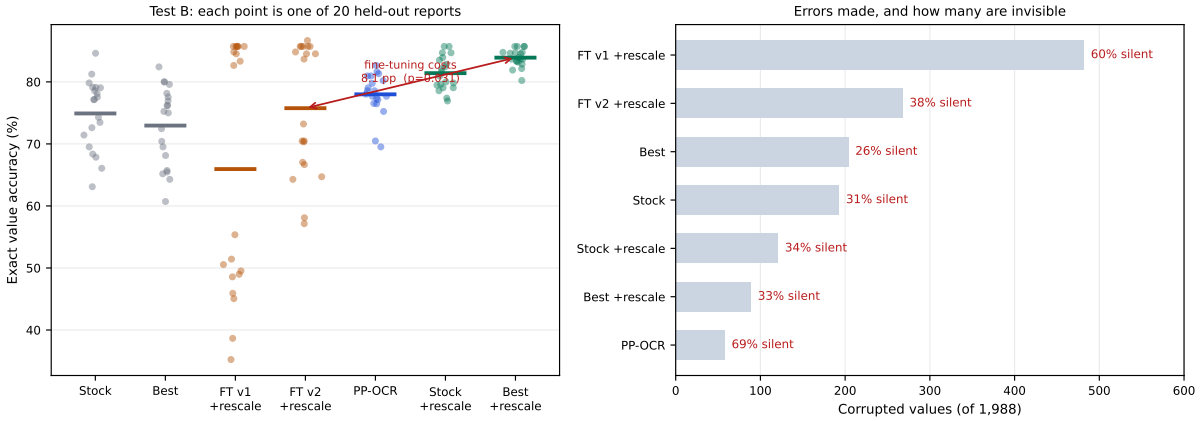


Figure 4: Left: exact value accuracy on each of the 20 held-out reports, with the mean marked. The fine-tuned model is both no more accurate on average than the stock model and far more variable between documents. Right: how many values each reader got wrong, annotated with the share of those errors that preserved the abnormality flag. PP-OCR makes the fewest errors and the largest proportion of its errors are invisible.

5.3 Analysis

Experiment 2 argued that analytes read from one page are not independent observations. The same argument applies here, so every comparison below treats the *document* as the experimental unit: exact value accuracy is computed within each of the 20 held-out reports and summarised across them, dispersion is the standard deviation between documents, and engines are compared by Wilcoxon signed-rank test over the 20 paired documents. Pooled counts are reported only for corruption totals, where the quantity of interest is the count itself.

5.4 Results

Table 5 reports exact value accuracy by reader, and Figure 4 shows the per-document distributions.

5.4.1 Fine-tuning is not the remedy it appears to be

Read only on Test A, the fine-tuned model is the clear winner: all 65 values exact in the resolution sweep at ≤150 dpi, where the stock pipeline reads 39 of 65 correctly (60.0%). That denominator deserves emphasis. The ≤150 dpi column covers five resolution conditions on one document, so 65 readings in

Table 6: Exact value accuracy on Test B by typeface, 10 reports each. Neither face appears in any fine-tuning corpus. Version 1 was trained on sans-serif faces only and version 2 added serif and monospace families; the fine-tuned models are the only readers whose accuracy depends strongly on which unseen face they meet, and the dependence shrinks as training typography widens.

Reader	Times	Courier	Difference
Stock Tesseract 5.5	71.0%	78.8%	−7.8
Tesseract best	67.8%	78.0%	−10.2
Fine-tuned v1 + rescale	85.0%	**46.8%**	**+38.2**
Fine-tuned v2 + rescale	**85.3%**	66.3%	+19.0
PP-OCR (RapidOCR)	78.4%	77.4%	+1.0
Stock + rescale	83.0%	79.9%	+3.1
Tesseract best + rescale	83.4%	84.4%	−1.0

total, all sharing a single rendering and layout. It is the most favourable comparison available to the fine-tuned model and the least independent.

On held-out reports it does not. Comparing it with the stock pipeline would confound two interventions, because the fine-tuned configuration is also rescaled, so we compare it against its own base model under identical preprocessing. Fine-tuned v2 with rescaling reaches 75.8% against 83.9% for `tessdata_best` with rescaling alone: fine-tuning costs 8.1 percentage points (Wilcoxon signed-rank over the 20 documents, $p = 0.031$). Version 1 under the same comparison costs 18.0 points ($p = 0.020$). Adding serif and monospace families to the training corpus recovered 9.8 points between the two versions ($p = 0.0015$), which narrowed the deficit without removing it.

Fine-tuning also produces more corrupted values than the base model it was built to improve (268 against 89), and a larger share of those corruptions are silent (37.7% against 32.6%).

The between-document standard deviation is the more revealing number. The fine-tuned model varies across documents with an SD of 10.5 percentage points, against 1.4 for the best rescaling configuration. It is not merely no better on average; it is unpredictable, which for a clinical pipeline is the worse property.

The failure is interpretable, and it is specific to typography. Test B is evenly split, 10 reports in Times and 10 in Courier, which separates the two cases (Table 6). The fine-tuned model is the best reader in the panel on unseen Times at 85.3%, and the worst on unseen Courier at 66.3%, against 84.4% for the base model it was built from under the same rescaling. It gains 1.9 percentage points on the serif face and loses 18.1 on the monospace one.

The mechanism is a word-segmentation artefact. In monospace text the inter-character gap around a decimal point is as wide as a space, and a model trained only on proportional faces reads Courier `7.8` as `7  .  8`, which the row parser truncates to 7.

Version 1 of the fine-tune, trained on sans-serif faces alone, exhibits this severely. Its result files had not been retained, so we re-ran it on Test B for this manuscript: it reads unseen Times at 85.0% and unseen Courier at 46.8%, a gap of 38.2 percentage points, producing 482 corrupted values of which 288 are silent. Version 2 added serif and monospace families to the training mix and halved the gap to 19.0 points, but did not remove it. The progression is the clearest statement of the problem: adding font diversity to the training corpus reduces the dependence on typography without eliminating it, while the configuration that never learned typography at all shows no dependence.

5.4.2 Upscaling the page is the single largest gain

Rescaling to at least 300 dpi before recognition requires no training, no new model and no change of engine. Each comparison below holds the recognition model fixed and varies only preprocessing. Over the 20 held-out documents, rescaling improves the stock model by 6.5 percentage points ($p = 0.0012$) and `tessdata_best` by 11.0 points ($p = 0.0001$). It takes low-resolution accuracy on Test A from 60.0% to 98.5% and reduces decimal-point losses on Test B from 109 to 11. The best configuration we

Table 7: Qwen2.5-VL 3B against the stock pipeline on identical cells. The Test B subset covers only three conditions, none of which is the poor-fax condition on which the model's characteristic failure was observed, so the 100% figure should not be read as a general result.

	Cells	Conditions	Qwen2.5-VL exact	Stock exact
Test A	260	12 of 64	85.0%	59.2%
Test B	339	3 of 7 (300 dpi, 100 dpi, phone photo)	100%	82.6%

tested is therefore the unmodified `tessdata_best` model with rescaling, which also carries the lowest between-document variability in the panel.

The reason is mechanical rather than statistical. A decimal point at 100 dpi occupies a handful of pixels and the recogniser's line model has little evidence to work with. Upscaling adds no information, but it presents the information already present at a scale the model was trained for. It also cannot overfit, because it introduces no parameters.

5.4.3 The vision–language model rewrites rather than reads

The vision–language model was run on a subset of conditions, so it is reported separately (Table 7) and compared against the stock pipeline on exactly the same document and condition cells.

Where the page is legible the model transcribes more accurately than any other reader tested, and it never dropped a decimal point. On fax-quality pages its behaviour changes in kind rather than degree. Instead of emitting a garbled token or no value, it produces fluent, well-formed values that are not printed on the page, and the substitutions move results toward the normal range: HbA1c 9.2 returned as "5.2 %, range 4.5–5.8, N"; haemoglobin 9.4 flagged *Low* returned as 13.4; potassium 5.8 flagged *High* returned as 3.6 with an invented unit and reference range. It also invented flags on clean pages and converted g/dL to g/L.

Every one of these passes a physiological plausibility check, because the values are physiologically ordinary. This is a worse failure mode than a dropped decimal point: a lost decimal produces an implausible number that a bound can catch, whereas an invented normal value is indistinguishable from a correct one by any check that inspects the value alone. The direction compounds the problem, since values are moved toward normality and therefore away from attention. Governance layers have been reported to reduce hallucination rates substantially in document extraction pipelines [9], which is consistent with our finding that the remedy is external verification rather than a better model.

5.5 A taxonomy of what actually goes wrong

Across both test sets the errors fall into five mechanisms.

1. **Lost decimal point.** At low resolution the separator disappears: creatinine 2.1 read as 21, white cell count 7.4 read as 74. The decimal in the printed reference range disappears with it ("0.6–1.2" becomes "06–12"), so value and range shift together and a row-local consistency check cannot detect the change.

2. **Range fall-through.** When the result cell is illegible, a parser that takes the first number on the line falls through to the lower bound of the reference range: INR 1.1 read as 0.8, glucose 77 read as 70.

3. **Digit substitution.** 31 read as 32, 195 as 198, 121 as 127. Plausible, same flag, invisible to any rule.

4. **Split decimal.** A monospace decimal point read as a word break, so `7.8` becomes `7 . 8` and the parser truncates to 7. Introduced by the fine-tuned models and absent from every stock configuration (Table 6).

5. **Generative rewriting.** Produced only by the vision–language model.

Mechanisms 1, 2 and 4 have structural signatures that a checker can exploit. Mechanisms 3 and 5 do not, and require a second opinion.

6 Experiment 4: what catches the errors that remain

No reader in Table 5 avoids undetected errors across all conditions. Experiment 4 therefore tests verification rather than recognition. Nothing is corrected automatically; a failed check routes the value to human review against the source image, which is the only safe action when the system cannot tell which of two readings is right.

Terms are used precisely below. A value is *corrupted* if the reader emitted a number that differs from truth. A check *catches* a value by routing it to human review; nothing is corrected automatically, and a caught value is not a corrected value. A *false alarm* is a correct value routed to review. The *review rate* is the share of all extracted values a human would inspect, and *undetected wrong* values are corrupted values the system accepted without review. The denominators differ by test set: 619 of 832 values were extracted at all on Test A, of which 43 were corrupted; 1,685 of 1,988 were extracted on Test B, of which 193 were corrupted. Values that were never extracted are absences the caller can observe and are excluded from these counts.

Four checks are layered on the existing plausibility bound. The *precision* check flags a value printed without a decimal for an analyte that laboratories always report with one. The *magnitude* check flags a value far above its plausible window where division by a power of ten would fit, restricted to the high side and to narrow-range analytes so that genuine extremes such as troponin in infarction are not flagged. The *structure* check detects range fall-through by testing whether the extracted number is immediately followed by a range separator and a second number. The *agreement* check requires a second, independently built reader to return the same value.

The claim that the two readers are independent is architectural rather than statistical, and it the distinction matters. Tesseract recognises a segmented text line with a recurrent LSTM over the whole line; PP-OCR detects text regions with a separate convolutional detector and recognises each crop independently, through a different training corpus and a different inference stack. They therefore fail on different inputs: Tesseract's characteristic error is losing a small separator within a line it has segmented correctly, while PP-OCR's is a detection boundary that clips or merges fields. Our data are consistent with this, since the two readers' error sets overlap only partially. What we have not established is statistical independence of their errors, which would require a failure model neither engine provides. Both could in principle fail identically on a sufficiently degraded page, and in that case agreement would confirm a wrong value rather than catch it. The check therefore reduces the chance of an undetected error; it does not eliminate it.

The magnitude check caught nothing the precision check had not already caught, on either test set, and we report that null result rather than omitting the row: the decimal losses it was designed to detect were all flagged first by the simpler precision rule. The remaining structural checks are nearly free, raising detection from 51.3% to 87.0% while increasing false alarms by 0.1 percentage points. Agreement with a structurally different reader closes the remainder, at the cost of routing roughly one value in five to review. With agreement enabled, every value the system accepted in both test sets was correct. The result holds when the fine-tuned model is the primary reader, where agreement still catches 266 of 268 wrong values on Test B.

The practical reading is that the second reader is doing work no single reader can do for itself, because the errors that survive the structural checks are precisely the ones that look normal.

The review burden translates as follows. At the structural tier, 10.7% of values are routed to review and 87.0% of wrong values are caught: for a clinic ingesting 100 reports a day with a 14-analyte panel, that is roughly 150 values to check against the source image. Adding the agreement check raises the catch rate to 100% and the burden to 19.8%, or roughly 280 values a day. Those are very different

Table 8: Verification performance, with stock Tesseract as the primary reader and PP-OCR as the second. *Wrong caught* is the number and share of corrupted values routed to review on Test B, with an exact binomial 95% confidence interval; *false alarms* is the share of correct values routed to review; *review rate* is the total share of values a human would inspect. Silent counts are from Test A. No clinician reviewed any output and no clinical decision was simulated: routing to review is a property of the software, not a measured improvement in care.

Checks	Silent caught · A	Wrong caught · B (of 193)	95% CI	False alarms · B	Review rate · B
None	0 / 28	0 (0.0%)	0.0–1.9	0.0%	0.0%
Plausibility only (existing)	9 / 28	99 (51.3%)	44.0–58.5	0.8%	6.6%
+ precision	23 / 28	158 (81.9%)	75.7–87.0	0.8%	10.1%
+ magnitude	23 / 28	158 (81.9%)	75.7–87.0	0.8%	10.1%
+ structure	23 / 28	168 (87.0%)	81.5–91.4	0.9%	10.7%
+ agreement with PP-OCR	**28 / 28**	**193 (100%)**	**98.1–100**	9.4%	19.8%

operational propositions, and the choice between them is not a technical one. It depends on document quality, on staffing, and on what the values are used for. What the data establish is the shape of the trade: the first 87% of errors are nearly free, and eliminating the remainder costs roughly a doubling of review volume.

7 Discussion

The practical consequence is that character-level accuracy does not bound value-level risk in a pipeline of this shape. A system reporting 88.9% character accuracy was delivering wrong values for roughly a third of the panel, and for nearly half of those the downstream alerting layer could not tell.

Threshold quantisation is common in clinical software, because reference ranges and severity bands are how laboratory results are presented. The finding here is that quantisation is also an information-destroying step placed immediately after an error-prone one. Any check that operates on the flag rather than the value inherits a blind spot whose size depends on how abnormal the true value was. The patients whose values are furthest outside the reference range are the patients whose corrupted readings are hardest to detect.

The plausibility bound is worth implementing and is not sufficient on its own. It is cheapest where it is least needed, for analytes like potassium whose physiological envelope is narrow, and it provides nothing for analytes like troponin whose plausible range spans several orders of magnitude.

One distinction in the results carries directly into how such a system should be read. An unread analyte is an absence the caller can notice; a misread one is a confident wrong answer. Pooling them, as a single extraction-failure rate would, conceals the difference between a failure that announces itself and one that does not.

Experiments 3 and 4 bear on how such a pipeline should be built. The instinct to fix a reading problem by training a better reader is, on our data, actively misleading. Evaluated on the document it was tuned for, the fine-tuned recogniser reads perfectly in the failure regime; set against its own base model under identical preprocessing, it is significantly worse on held-out reports and produces three times as many corrupted values. A practitioner who held out no unseen typography would have concluded the problem was solved and deployed a model that fails differently rather than less.

By contrast, the intervention that worked is not a model at all. Upscaling the page before recognition improves low-resolution accuracy more than any model change we tested, requires no training data, and cannot overfit, because it adds no parameters. When a failure is caused by information being presented at the wrong scale, the appropriate remedy is to change the scale.

The vision–language result deserves separate emphasis for clinical readers. Such models are increasingly proposed for document understanding, and on legible pages this one was the most accurate reader we tested. Its failure mode on illegible pages is not noise but confabulation, and the invented values

moved toward the normal range. A system that silently normalises abnormal results is more dangerous than one that fails loudly, and conventional accuracy reporting would have recorded this model as the best performer.

The pattern across these experiments is that undetected error is reduced by disagreement rather than by accuracy. Two readers built on different principles fail on different inputs, and requiring them to agree converts an undetectable error into a detectable one. On our data that reduced undetected wrong values to zero, at a review burden of roughly one value in five. Whether that burden is acceptable is a deployment question that depends on volume and on what the values are used for.

8 What follows for a system that reads clinical documents

The findings suggest five things a team building such a pipeline could act on. We state them as implications of the mechanism rather than as validated interventions, since none has been tested in a clinical setting.

1. **Do not report character accuracy as a safety figure.** It did not bound value-level error anywhere in our data, and the gap was widest in the degradation regime real documents occupy. Report the share of target values extracted exactly, which is the quantity the clinical logic consumes.

2. **Normalise resolution before recognition.** Upscaling to at least 300 dpi gave a larger improvement than any model change we tested, adds no parameters and cannot overfit. It is a few lines of code at the front of the pipeline.

3. **Hold out typography before trusting a fine-tuned recogniser.** Evaluated only on familiar documents, our fine-tuned model looked like a solution. On fonts it had never seen, measured against its own base model under the same preprocessing, it was significantly worse and produced more invisible errors. A held-out set in a deliberately different typeface separates those two conclusions and costs a few hours to build.

4. **Treat a generative reader's output as a claim, not a transcription.** The vision–language model was the most accurate reader on legible pages and, on illegible ones, produced plausible values that were not printed, biased toward the normal range. Any pipeline using such a model needs its output checked against a deterministic reader on the same region, because the failure is invisible to checks that inspect the value alone.

5. **Carry the distance, not just the side, of a threshold.** The flag is where the information is lost. A system that retains how far a value lies outside its reference range, or that compares against the range printed on the document rather than a hardcoded one, can detect errors that a flag cannot represent.

Of these, the second and third cost almost nothing and we would expect them to generalise. The fifth is a design change with wider consequences and would need evaluating in context.

9 Limitations

The documents are synthetic. They carry exact ground truth, which is what makes the measurement possible, and they are not real clinical documents. Nothing here establishes a rate for any deployed system, and no clinician reviewed any output.

One OCR engine was used, at one page segmentation mode, with one document layout and one panel of 13 analytes. Every quantity we report is conditional on that stack.

Resolution and pixels per glyph are entangled, since body text varies between 8.5 and 10 point across the generated documents. A reader could reasonably hold that character size in pixels, rather than nominal resolution, is the operative variable, and this design cannot separate them.

The plausibility envelope is our construction. Its zero false-positive rate is an artefact of a generator that samples inside it.

The degradation sweep in Experiment 1 and the extractor comparison use a single document. Only the resolution sweep is replicated across 10 documents, and only the resolution results carry between-document dispersion.

We did not measure whether a corrupted value changes a clinical recommendation. The propagation of a corrupted value into generated text was observed on one document and is not quantified here.

The held-out set in Experiment 3 is held out with respect to typography, layout and values, but not with respect to the generator. Test B and the fine-tuning corpus are produced by the same code and therefore share an analyte catalogue, a set of reference ranges, and the conventions by which numbers and ranges are formatted. A fine-tuned model could in principle exploit those shared conventions in a way it could not exploit a real laboratory's. This makes our held-out result conservative in one direction, since the fine-tuned model failed to generalise even on material that shares its generator, and optimistic in another, since we cannot measure how it would behave on genuinely foreign report conventions. Test B is also still synthetic and rendered digitally before degradation. Real scans add skew, shadows, staples, stamps, handwritten annotations and photocopying artefacts that our degradation model does not reproduce. The held-out comparison therefore tests generalisation across fonts and layouts, not across capture devices.

One vision–language model was evaluated, at 3 billion parameters, running locally, on a subset of conditions. Larger models read more accurately. We cannot say from our data whether the rewriting behaviour diminishes with scale, and we did not test that.

The analyte conventions underlying the precision and magnitude checks, meaning which tests are conventionally printed with a decimal and what counts as a narrow range, are typical adult values assembled for this study rather than a clinical standard. A deployment would take them from the laboratory's own information system.

We report four paired engine comparisons and apply no correction for multiple testing. Under a Bonferroni threshold of $\alpha = 0.0125$ the two rescaling results ($p = 0.0012$, $p = 0.0001$) and the comparison between fine-tuned versions ($p = 0.0015$) remain significant, while the two comparisons of a fine-tuned model against its base ($p = 0.031$ and $p = 0.020$) would not. The direction of those two is consistent across both model versions and with the per-typeface results, but a reader who requires correction for multiplicity should treat the magnitude of the fine-tuning penalty as less firmly established than its direction.

The confidence intervals on verification recall are exact binomial intervals over pooled corruptions and do not account for clustering within documents; they are descriptive rather than inferential. The pooled risk ratio has the same limitation, which is why the document-level paired analysis is reported alongside it and is the basis for the claim.

The review rate reported for the agreement check is measured on our degradation distribution. A document population with better average capture quality would produce fewer disagreements and a lower burden, and a worse one would produce more.

10 Conclusion

Character accuracy and value accuracy come apart under degradation, and the gap is widest exactly where the clinical stakes are highest. Nearly half of the value corruptions we observed preserved the abnormality flag, and corruption of an already-abnormal value was 2.80 times as likely to be silent as corruption of a normal one (95% CI 1.75–4.47), because a threshold cannot represent how far past it a value has moved. A plausibility bound recovers some of this and fails systematically for the analytes with the widest physiological ranges. Substituting a language model for a deterministic parser changed nothing, because the information was lost before either could read it. Knowing where a number came from is not the same as knowing it was read correctly.

Attempting to fix this by training a better recogniser produced a model that reads the document it

was tuned for perfectly and unseen typography significantly worse than the base model it was built from, while increasing the number of corruptions the alerting layer cannot see. Upscaling the page before recognition, which adds no parameters and cannot overfit, did more for low-resolution accuracy than any model we trained. A vision–language model read legible pages better than anything else and, on illegible pages, invented plausible values in the direction of normality. What most reduced undetected errors was none of these: it was requiring two independently built readers to agree, which left no undetected wrong value in either test set, within the limits that a finite sample with zero observed failures can establish. In a pipeline that reduces measurements to flags, the useful question is not how accurately the page was read, but whether anything downstream would notice if it had not been.

Data and code availability

All materials are available at `https://github.com/thejabirhussain/silent-corruption-clinical-ai`, released under an open licence. The repository contains the document generator and the truth files it writes, the degradation and recognition harness, the fine-tuning scripts and training logs for both model versions, the per-reading result tables from which every number in this paper is computed, the verification implementation, and the scripts that produce each figure and table. Reproduction instructions, including the exact commands for generating the held-out test set, training a model and evaluating any reader in Table 5, are in the repository README.

The fine-tuned Tesseract models are 15 MB each and are released with the code. Third-party components are obtained from their own distributions rather than redistributed: the stock Tesseract models from the `tessdata` and `tessdata_best` repositories, PP-OCR through `rapidocr_onnxruntime`, and Qwen2.5-VL 3B through Ollama.

Synthetic data are used throughout. No real patient information was processed at any stage, and no component of the released material contains or derives from patient data.

Ethics approval

This study required no ethical approval and none was sought. It uses only synthetic laboratory reports generated programmatically for this work. No human subjects, human tissue, identifiable personal data or real clinical records were involved at any stage, and no participant identifying information appears in the manuscript or in the released materials.

Author contributions

M.J.H.S. is the sole author. He conceived the study, designed and implemented the experiments, generated the synthetic corpora, fine-tuned and evaluated the recognition models, implemented the verification layer, performed the statistical analysis, produced the figures and wrote the manuscript. The clinical decision support prototype whose extraction path is measured here was also built by the author, as an unfunded academic project.

Funding

This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.

Competing interests

The author declares no competing interests. Neither the author nor the author's institution has received payments or services from any third party related to this work in the past 36 months. The clinical decision

support prototype measured in this study was built by the author as an unfunded academic project and is not a commercial product.

Use of artificial intelligence

The author used Anthropic Claude as a coding assistant, to help write and debug the experiment and analysis code, and to help draft and check the manuscript text. The research question, experimental design, execution, analysis and interpretation are the author's own. The author reviewed and edited all content, verified every reported value against the released result files, and takes full responsibility for this manuscript.

References

[1] Amugongo, L. M., et al. (2025). Retrieval augmented generation for large language models in healthcare: a systematic review. *PLOS Digital Health*, 4(6), e0000877.

[2] Ben Abacha, A., and Demner-Fushman, D. (2019). A question-entailment approach to question answering. *BMC Bioinformatics*, 20, 511.

[3] Yu, A., Weile, J., and Courtot, M. (2026). Benchmarking LLM-based information extraction tools for medical documents. *medRxiv*, 2026.01.19.26344287. https://doi.org/10.64898/2026.01.19.26344287

[4] Collins, G. S., Moons, K. G. M., Dhiman, P., et al. (2024). TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ*, 385, e078378.

[5] Gao, Y., et al. (2023). Retrieval-augmented generation for large language models: a survey. *arXiv:2312.10997*.

[6] Kilickaya, O., Schmickl, C., Ahmed, A., Pulido, J., Onigkeit, J., Kashani, K., Gajic, O., Herasevich, V., and Pickering, B. (2014). Customized reference ranges for laboratory values decrease false positive alerts in intensive care unit patients. *PLoS ONE*, 9(9), e107930. https://doi.org/10.1371/journal.pone.0107930

[7] Hemmer, A., Coustaty, M., Bartolo, N., and Ogier, J.-M. (2024). Confidence-aware document OCR error detection. *arXiv:2409.04117*.

[8] Lu, Y., Zhao, X., and Wang, J. (2024). ClinicalRAG: enhancing clinical decision support through heterogeneous knowledge retrieval. In *Proceedings of the KnowLLM Workshop, ACL 2024*, pp. 64–68.

[9] Sediqin, M., Dalmia, S., Thoppanahalli, S., Kovvuri, S. K. R., and Mukherji, A. (2026). SAGE: governed artifact generation from enterprise guidelines. *arXiv:2609.17775*.

[10] Smith, R. (2007). An overview of the Tesseract OCR engine. In *Ninth International Conference on Document Analysis and Recognition (ICDAR)*, pp. 629–633.

[11] Wang, Y., Li, Y., Qin, Y., and Qian, H. (2026). Key coverage matters: semi-structured extraction of OCR clinical reports. *arXiv:2605.09440*.